



AI in Medicine Series: Final Assignment

| INFORMATION

Time to apply what you've learned and do your own investigating!

The goal of this assignment is for you to develop an understanding of AI implementations from a collaborator/stakeholder perspective. This assignment has four sections covering (1) model cards, (2) data cards, (3) applications to medicine, and (4) utilizing general LLM models.

We are releasing this now so that you may start reading through the excersizes and have plenty of time to ask questions. Cramming this right before the deadline is ill advised.

Getting Help

We highly encourage you to discuss and solidify ideas with your peers! However, plagiarism is always unacceptable and will be dealt with accordingly. Questions may be asked at the end of the sessions as well as on a per email basis.

For Assignment/Technical Help:

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Deadlines

Part 3 requires a declaration of topic that needs to be pre-completed via Google Form by January 19th, 2026.

**All 4 sections must be completed by April 5th, 2026 to receive credit.
Have fun!**

| 1. MODEL CARDS

Read about model cards here: [Google Model Cards](#)

Play with Google's healthcare models here: [Demos](#)

1. Choose 1 model, and answer the following questions:

Intended Use

- What problem is this model designed to solve?

Training Data

- What kinds of data were used?
- Are they representative of all populations?

Limitations

- What groups might this model not work well for?

Computation

- How much computation was used for building the model?
- How might the model's computational requirements affect its use in low-resource clinical settings?

2. Pick two or three models, and answer the following:

- How do the goals of these two models differ?
- Which one can be applied to a broader range of problems?

3. Imagine your daily experience as a medical student, and in the future as a physician:

- What are the challenges of incorporating one of these models as a tool into the current workflow and tools available for diagnosis?
- What additional information would you want in a model card before relying on the model to make decisions with?
- Imagine a scenario where a patient asks you: "How does this AI tool know I have this condition?" How would you explain it in plain language?
- How could bias in training data harm underrepresented patient populations?
- What safety considerations should be taken into account before deploying this tool in a hospital or clinic? (continuous monitoring, validation phases, etc.)

| 2. DATA CARDS

Pick a dataset and build a data card.

Potential data sources you can use:

- [VitalDB](#)
- [Kaggle Top Medical Datasets](#)
- [MedMNIST](#)
- [Alzheimer MRI Disease Classification Dataset](#)
- [Breast Cancer Dataset](#)
- [Mental Health Dataset](#)
- [Chronic Kidney Disease dataset](#)
- [Heart Disease Dataset](#)

1. What healthcare problems are you interested in solving? Find a related dataset, and build a data card. Focus on questions like:

- Is it classification or regression?
- How is the data distribution?
- Are there any imbalances? Missing values? Inherited bias in the data?
- Is there a feature or characteristic that should be collected in the dataset, but it's not?
- What features are more important than the others based on clinical knowledge?

2. Read the template for datasheets for healthcare from here: [The Data Cards Playbook and Templates](#)

- Which questions are of more importance in your area of interest?
- Which questions play a less important role?
- How can these questions help build a roadmap for your research on the dataset you have and scope your research project?

| 3. ASK A RESEARCH QUESTION

Read a few sample model cards and data cards and think about how they collect data and why they use the models they use. Then think of how you can apply them to the problems you're interested in.

1. Submit a topic of interest.

A Google Form will be open from January 12 to January 19, 2026.

Submit a topic of interest that you will address in your proposal.

Some ideas may be:

- AI triaging in emergency medicine
- Anomaly detection of rare pathologies
- AI-assisted clinical documentation
- Risk stratification of heart failure
- Causal ML for treatment response

2. Write a short research proposal on your proposed topic. Focus on data cards and model cards.

| 4. TRY AN LLM

Pick a large language model and try it in your workflow.

Potential LLMs you can use:

- [NotebookLM](#) (Google models are university approved)
- [ChatGPT](#)
- [Grammarly](#)
- [Claude](#)
- [Perplexity](#)
- [Llama](#) (requires coding setup)
- [DeepSeek](#)

1. Copy and paste your report into an LLM and prompt it to edit/improve it. In your submission, include:

- What LLM you used
- Each prompt you used, and
- The results from each prompt

2. Reflect on how the LLM performed. Focus on:

- What was improved in your report
- What was worsened in your report
- Overall impression of the edited report
- Overall impression of the process

We will not be grading/penalizing you on your prompting, we will only grade your reflection in question 2.

| EXTRA RESOURCES

[Demos and Notebooks - Spark NLP](#)

Try extracting medical entities from a sample note. Which outputs are useful, and which are confusing or misleading?

[Microsoft Healthcare AI](#)

Go to one of the demos, e.g. chest x-ray report generation. Compare the result of the model with your knowledge of the features present in the X-ray.

[Open Medical-LLM Leaderboard](#)

Pick one LLM from the Open Medical-LLM Leaderboard. What tasks were it evaluated on? How does it compare to general-purpose models like ChatGPT? What are your thoughts about the benchmarks used? Are they reliable, or can there be biases in them?

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